



OHDSI / OMOP Environment Setup

Prerequisites and installation guide for participants

NOTE

Most of this setup takes about 45 minutes. Two steps take longer and are worth starting early: the Athena vocabulary package is prepared on Athena's servers and emailed to you, and the first Usagi vocabulary index can take several hours to build.

1. What you are installing, and why

COMPONENT	WHAT IT DOES	PLATFORM
White Rabbit	Scans your source tables, fields and values and produces a scan report.	Windows / macOS
Rabbit-in-a-Hat	Designs the source-to-CDM mapping and documents the ETL. Ships inside White Rabbit.	Windows / macOS
Athena	Web service for browsing the Standardized Vocabularies and downloading vocabulary files.	Browser — nothing to install
Usagi	Proposes candidate Standard Concepts for your source codes; you review and approve them.	Windows / macOS
PostgreSQL	Database that holds the OMOP CDM and the vocabulary tables.	Windows / macOS
pgAdmin 4	Graphical client for PostgreSQL — connecting, browsing, running SQL.	Windows / macOS
R	Statistical environment the OHDSI analytical packages run in.	Windows / macOS
RStudio Desktop	Editor you will actually work in. Requires R to be installed first.	Windows / macOS
Rtools	Compiler toolchain R needs when a package must be built from source.	Windows only

THREE THINGS PEOPLE GET WRONG

- Athena is a web service, not an application. There is nothing to install.
- Rabbit-in-a-Hat is not a separate download — it is inside the White Rabbit package.
- Rtools is Windows-only. macOS participants should not install it.

2. Before you start

2.1 Hardware and connectivity

- 8 GB RAM minimum. 16 GB strongly preferred — the Usagi index build is memory-hungry.
- At least 25 GB free disk space for the vocabulary files and the index.
- A stable connection for the downloads. Do not plan to download anything large at the venue.

2.2 Java

White Rabbit, Rabbit-in-a-Hat and Usagi are Java applications. Install Java 8 for this training.

Download: java.com/en/download

- **Windows:** use the Windows installer offered on that page.

- **macOS:** choose the build matching your Mac. Oracle publishes both an **ARM64** build for Apple Silicon (M1–M4) and an **x86_64** build for Intel Macs.

Not sure which Mac you have? Run this in Terminal — it returns arm64 or x86_64:

```
| uname -m
```

Then confirm Java is installed, on either platform:

```
| java -version
```

You should see a version beginning 1.8. Command Prompt on Windows, Terminal on macOS.

IF YOU ALREADY HAVE A NEWER JAVA

You do not have to remove it. The OHDSI tools accept Java 8 or higher. Install Java 8 only if java -version reports nothing, or reports a version the trainers have told you is unsupported.

2.3 Athena account

Register or sign in at athena.ohdsi.org. Browsing concepts works without an account; downloading vocabulary files does not.

3. OHDSI tools

3.1 White Rabbit and Rabbit-in-a-Hat

1. Download the latest release: github.com/OHDSI/WhiteRabbit/releases/latest
2. Unzip it somewhere you can find again. Avoid folder names with spaces or accented characters.
3. Launch:
 - **Windows:** bin\whiteRabbit.bat or bin\rabbitInAHat.bat
 - **macOS:** bin/whiteRabbit or bin/rabbitInAHat from Terminal

macOS: FIRST LAUNCH MAY BE BLOCKED

macOS blocks unsigned applications by default. Go to System Settings → Privacy & Security, find the blocked item near the bottom, and choose Open Anyway. This affects White Rabbit and Usagi.

3.2 Usagi

1. Download Usagi_vx.x.x.jar from github.com/OHDSI/Usagi/releases
2. Start it from the command line. This is more reliable than double-clicking, and it shows you any error:


```
java -jar Usagi_vx.x.x.jar
```
3. On first start, Usagi asks where your unzipped vocabulary files are and builds its index.

THE INDEX BUILD IS THE LONG STEP — DO IT EARLY

- It can run for several hours. It is done once.
- It needs the Athena files first, so request those before you start.
- If it appears frozen after a few hours, close it and relaunch from the command line as above — this is a documented issue, not your computer.
- When it finishes, the vocabulary version appears at the bottom-right of the Usagi window. That is your confirmation.

3.3 Vocabulary files from Athena

1. Sign in at athena.ohdsi.org and open the Download tab.
2. Select the vocabularies the trainers specify. Typically SNOMED CT, LOINC, RxNorm and RxNorm Extension, source vocabularies such as ICD10 / ICD10CM, and the OMOP-generated vocabularies.
3. Submit the request. Athena builds the package and emails you a download link.
4. Download and unzip. Keep the folder — this is what you point Usagi at.

4. Database environment

4.1 PostgreSQL and pgAdmin 4

Download the installer for your platform: postgresql.org/download. The standard EDB installer includes pgAdmin 4.

- Install both PostgreSQL Server and pgAdmin 4 when the installer offers the component list.
- Set a password for the postgres user and write it down. You cannot recover it later.
- Keep port 5432 unless it is already in use on your machine.

If pgAdmin was not installed, get it separately from pgadmin.org/download.

Test the connection in pgAdmin using host localhost, port 5432, database and user postgres.

5. R analysis environment

Install in this order. RStudio looks for an existing R installation, so R must come first.

5.1 R

Install from CRAN: cran.r-project.org. Choose "Download R for Windows" or "Download R for macOS".

- **macOS:** CRAN publishes separate builds for Apple Silicon and Intel. Pick the one matching the result of `uname -m` from section 2.2.
- Install the release version your trainers specify, not necessarily the newest.

5.2 RStudio Desktop

Install the free desktop edition: posit.co/download/rstudio-desktop

Open RStudio and run this in the Console:

```
R.version.string
```

If it prints a version, RStudio can see R and both are working.

5.3 Rtools — Windows only

Install the Rtools version matching your R version: cran.r-project.org/bin/windows/Rtools

The versions must correspond — Rtools 4.4 goes with R 4.4.x, and so on. Verify in the R console:

```
Sys.which("make")
```

A file path confirms Rtools is installed and visible to R. An empty result means it is not.

macOS PARTICIPANTS

Do not install Rtools — it does not exist for macOS. If a package later needs compiling, the trainers will handle the macOS toolchain during that session.

6. Readiness checklist

Run each test yourself. If one fails, photograph or copy the error message and send it to the training team before you travel.

	ITEM	TEST — AND WHAT A PASS LOOKS LIKE
☐	Java	<code>java -version</code> returns a version starting 1.8
☐	White Rabbit	The application window opens
☐	Rabbit-in-a-Hat	The application window opens
☐	Athena account	You can sign in at athena.ohdsi.org
☐	Vocabulary files	<code>CONCEPT.csv</code> is present in the unzipped folder
☐	Usagi	The application window opens
☐	Usagi index	The vocabulary version shows at the bottom-right of the Usagi window
☐	PostgreSQL	The service is running and accepts your postgres password
☐	pgAdmin 4	Connects to localhost:5432 as user postgres
☐	R	<code>R.version.string</code> prints a version in the RStudio Console
☐	RStudio Desktop	Opens, and the Console responds
☐	Rtools (Windows)	<code>Sys.which("make")</code> returns a file path
☐	Training data	You have the dataset, the data dictionary and the source-code file

7. Where to get help

OHDSI forums — the people who build these tools answer questions there: forums.ohdsi.org

The Book of OHDSI, the reference text: ohdsi.github.io/TheBookOfOhdsi

OMOP CDM specification: ohdsi.github.io/CommonDataModel

Vendor pages: java.com · cran.r-project.org · posit.co · postgresql.org · pgadmin.org

READY?

If every applicable line in section 6 passes, your machine is ready for the practical. Bring the laptop you tested on, and bring your postgres password.